

Software FPU Addition Algorithm

Haris Shibli (28988)

16-BIT IMPLEMENTATION

DATA TYPE

Custom 16-bit Float

$$V = (-1)^S \times 1.M \times 2^{(E-15)}$$

SIGN

1 bit

EXPONENT (Bias = 15)

5 bits

MANTISSA

10 bits

Bias 15 · equivalent to IEEE-754 half (binary16) · normal range $\approx 6.1 \times 10^{-5} - 6.5 \times 10^4$ · ~3 significant digits

01

Extraction

AND field-masks split **S** / **E** / **M**. Prepend the hidden 1 → 11-bit significand (1.M) for the ALU.

AND · OR

02

Alignment

Subtract exponents for Δe . Barrel-shift the **smaller significand right** by Δe ; keep the dropped MSB as a guard bit.

SUB · SHR

03

Arithmetic

Signs equal → ADD. Signs differ → **two's-complement** smaller (XOR+1), ADD; result sign follows the larger operand.

XOR · ADD

04

Normalization

Restore 1.XXX. Carry-out → **SHR**, **E+1**. Leading zeros → **SHL**, **E-1** each, until the hidden 1 returns.

SHL · SHR

05

Packing

Drop hidden 1, clamp E to 5 bits, **SHL** each field into its slot, **OR** into one 16-bit TOY word.

SHL · OR

Execution Trace: Add 6.0 + 2.5

Input N1: 0 10001 1000000000 (6.0)

Input N2: 0 10000 0100000000 (2.5)

S1 Extract: M1=1.100 (1.5) M2=1.010 (1.25)

S2 Align $\Delta e=1$: M2»1=0.101 (0.625) E=17

S3 Add: 1.100+0.101=10.001 (2.125)

S4 Norm(ovf): 1.0001 (1.0625) E=18

S5 Packed Result (8.5)

0 10010 0001000000

SPECIAL CASES & ROUNDING

- Zero E = M = 0 (hidden 1 suppressed)
- Overflow E > 30 → saturate to $\pm\max$
- Underflow E < 1 → flush to ± 0
- Rounding guard bit, round-half-up